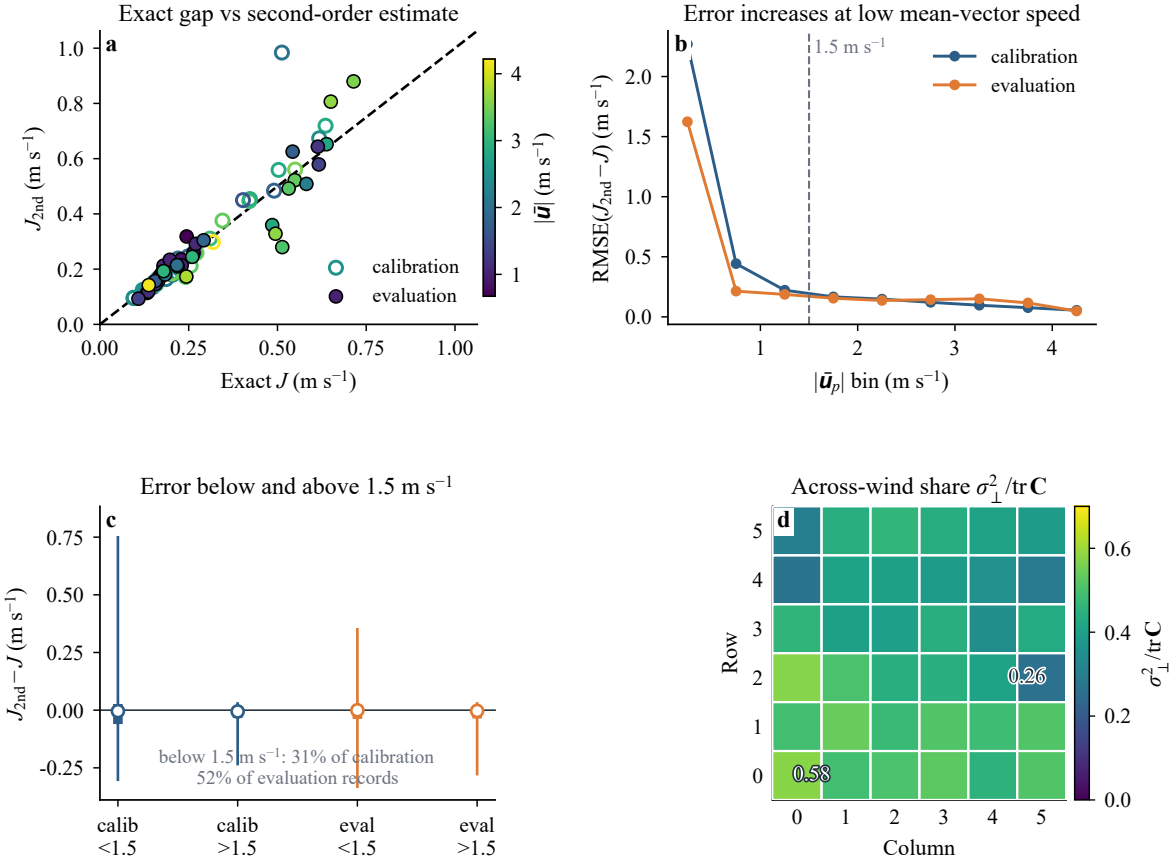

Exact gap vs second-order estimate
a
J_{2nd} (m s^{-1})
Exact J (m s^{-1})
calibration
evaluation
$|\bar{\boldsymbol{u}}|$ (m s^{-1})
Error increases at low mean-vector speed
b
RMSE($J_{2nd} - J$) (m s^{-1})
1.5 m s^{-1}
calibration
evaluation
$|\bar{\boldsymbol{u}}_p|$ bin (m s^{-1})
Error below and above 1.5 m s^{-1}
c
$J_{2nd} - J$ (m s^{-1})
below 1.5 m s^{-1}: 31% of calibration
52% of evaluation records
calib <1.5
calib >1.5
eval <1.5
eval >1.5
Across-wind share $\sigma_\perp^2/\mathrm{tr}\,\mathbf{C}$
d
Row
Column
0.26
0.58
$\sigma_\perp^2/\mathrm{tr}\,\mathbf{C}$